LEARNER GUIDE

Welcome to

Learning Footprint Labs

Your hands-on lab environment runs fully in your web browser — nothing to install on your computer. This guide walks you through everything, step by step.

https://labs.learningfootprints.com

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Overview

What you will be using

Learning Footprint Labs gives you a personal, ready-to-use computer — your "lab environment" — that opens right inside your browser. It already has all the tools and your course materials installed, so you can start practising immediately, from any laptop or desktop, on any operating system.

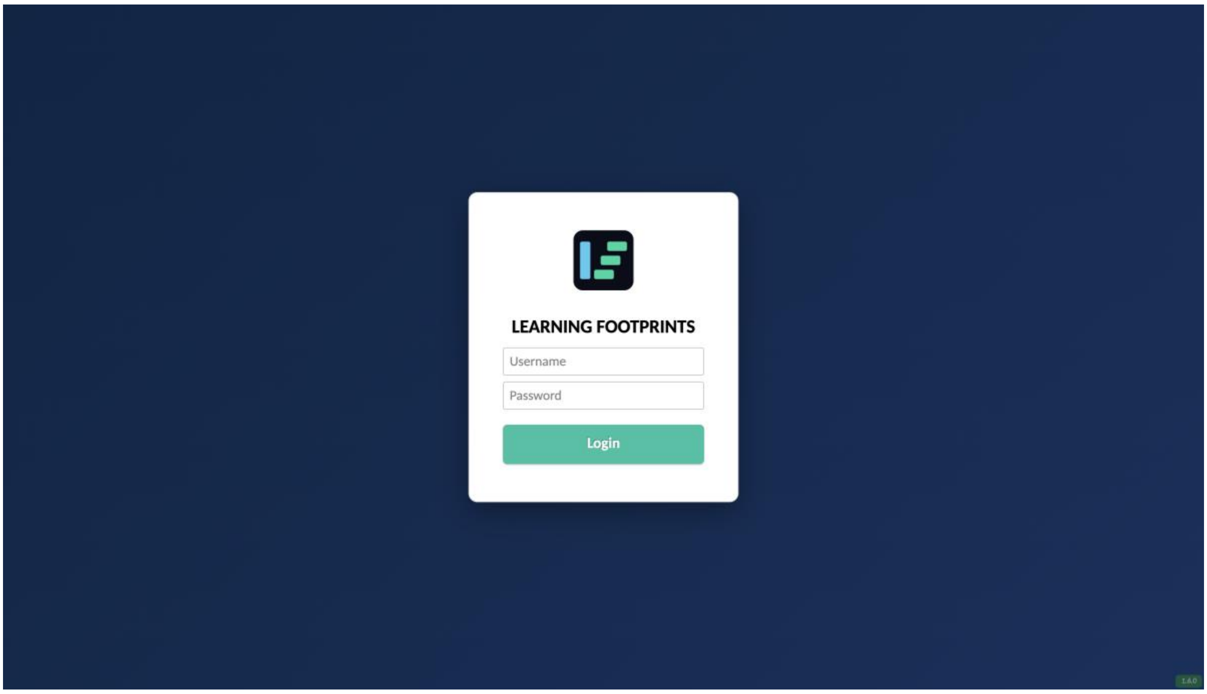
Before you begin — what you need

- A modern web browser (Google Chrome, Microsoft Edge, Firefox, or Safari).
- A stable internet connection.
- Your personal **login details** (email address and password), given to you by your program coordinator.

Step 1 — Open the labs website

Open your web browser and go to: **https://labs.learningfootprints.com**

You will reach the Learning Footprint Labs sign-in page. We suggest you bookmark this address so you can come back to it quickly.



The Learning Footprint Labs sign-in page.

✓ Tip — check it is secure

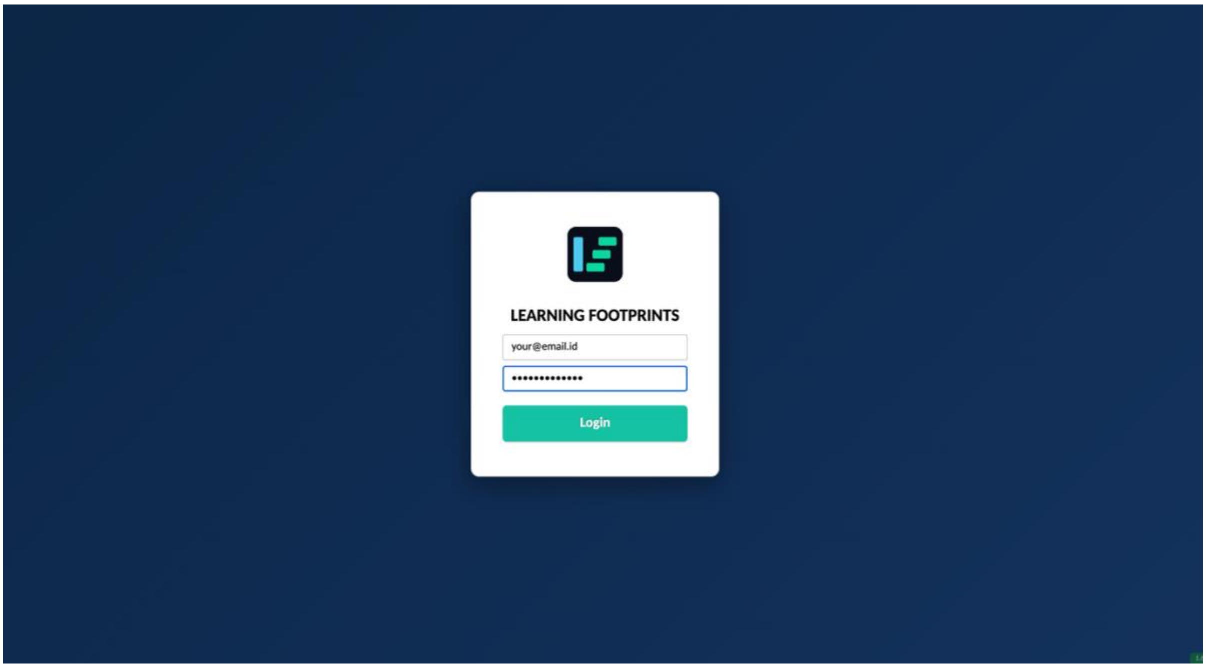
Before you type your password, check that the address bar shows **https://** and a padlock icon. This confirms you are on the genuine, secure labs site.

Step 2 — Sign in

Type the details you were given:

- **Username** — your registered email address (for example, yourname@yourorg.com).
- **Password** — the password issued to you.

Then click **Login**.



Type your email and password, then click Login.

⚠ Keep your login private

Your login is personal and tied to your own lab. Please do not share it. If you are asked to change your password on first login, choose a strong one (at least 12 characters, mixing upper- and lower-case letters, a number, and a symbol).

Step 3 — Your lab environment

After you sign in, Learning Footprint Labs takes you to your **assigned lab environment** — a complete desktop computer prepared just for your course.

Most learners are assigned a **single environment**. In that case you are taken **straight into your lab** — there is nothing extra to click. If you have been assigned **more than one** environment, you will first see a short list; simply click the name of the environment you want to open.

One machine, or several?

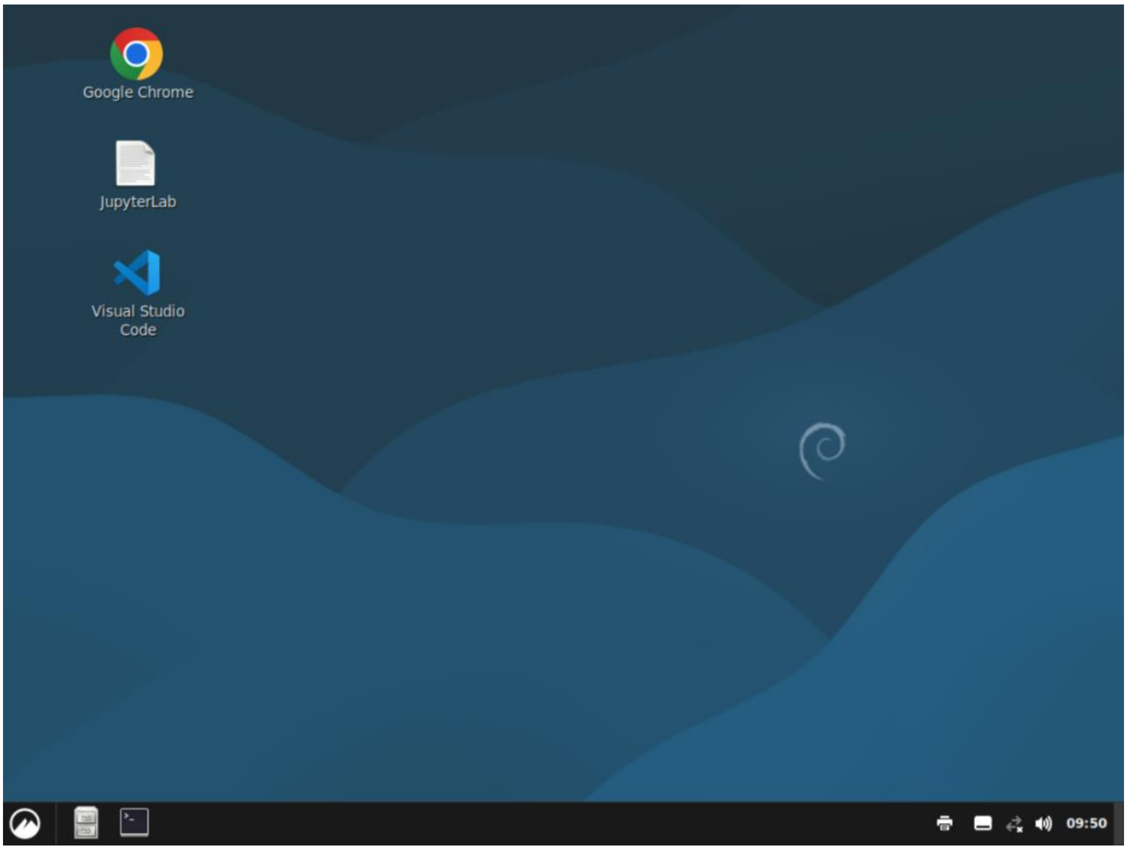
If you have one machine assigned — you are taken directly into it, straight to the lab desktop.
If you have more than one machine assigned — you first see a list of your machines. Click the name of the machine you want, and it opens.

What is an "environment"?

Think of it as your own dedicated machine in the cloud. It stays set up between sessions, so the work you save inside it will be there when you return.

Step 4 — Open your lab

Your lab desktop loads and fills the browser window — this is your live, interactive computer. Give it a few moments to connect the first time.



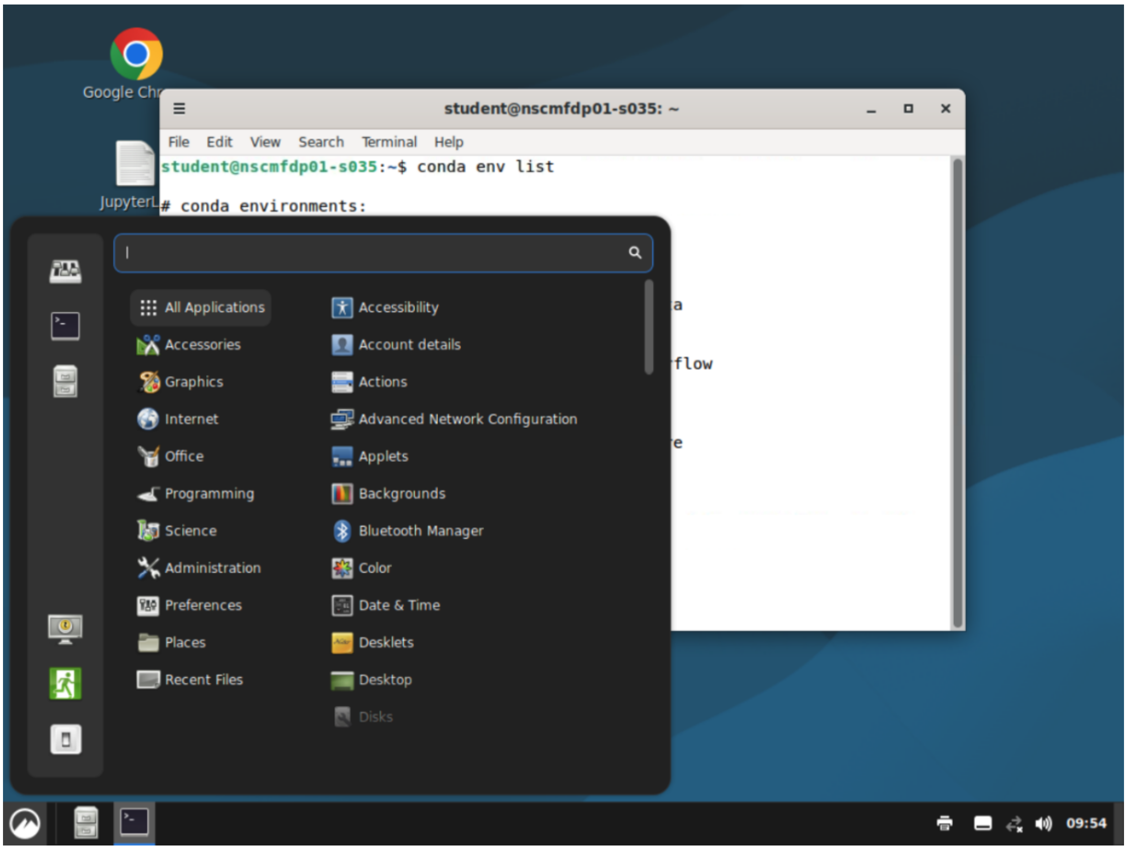
Your lab desktop, running fully inside the browser.

> **✓ Tip — go full screen**
> For the best experience, put your browser into full-screen mode — press **F11** on Windows or Linux, or **Ctrl + Cmd + F** on a Mac — so the lab uses your whole display.

Step 5 — Inside the lab desktop

You are now working on your lab computer, just like a normal desktop. Your environment comes pre-loaded with all the tools and learning setups for your programme.

At the bottom-left of the screen is the **applications menu**. Click it to see everything that is installed, grouped into categories such as **Programming**, **Science**, **Internet**, and **Office**.



The applications menu, showing the installed software by category.

Desktop tools that are installed

Your lab desktop already includes everything you need for the course:

Tool	What you use it for
JupyterLab	Write and run Python notebooks. One JupyterLab serves all the conda environments listed below, so you can pick the right kernel for each topic. Open it from the desktop icon.
Visual Studio Code	A full code editor with the Python and Jupyter extensions already installed — ideal for writing and debugging code. Open it from the desktop icon.
Google Chrome	For documentation, research, and any online resources your course needs.
Terminal	A command line for running commands, managing conda environments, and using git. Open it from the taskbar at the bottom.
Files	The file manager, for browsing your home folder and your courseware (see Step 8). Open it from the taskbar.
LibreOffice + viewers	Open and edit Word, Excel, and PowerPoint files, and read PDFs.
Docker, git & data tools	Docker and Compose, git, the AWS CLI, and database clients (redis-tools, mongosh) are pre-installed for hands-on exercises.

Step 6 — Adjusting your screen (size and resolution)

If the lab looks too small or too large, you can fix it in two ways.

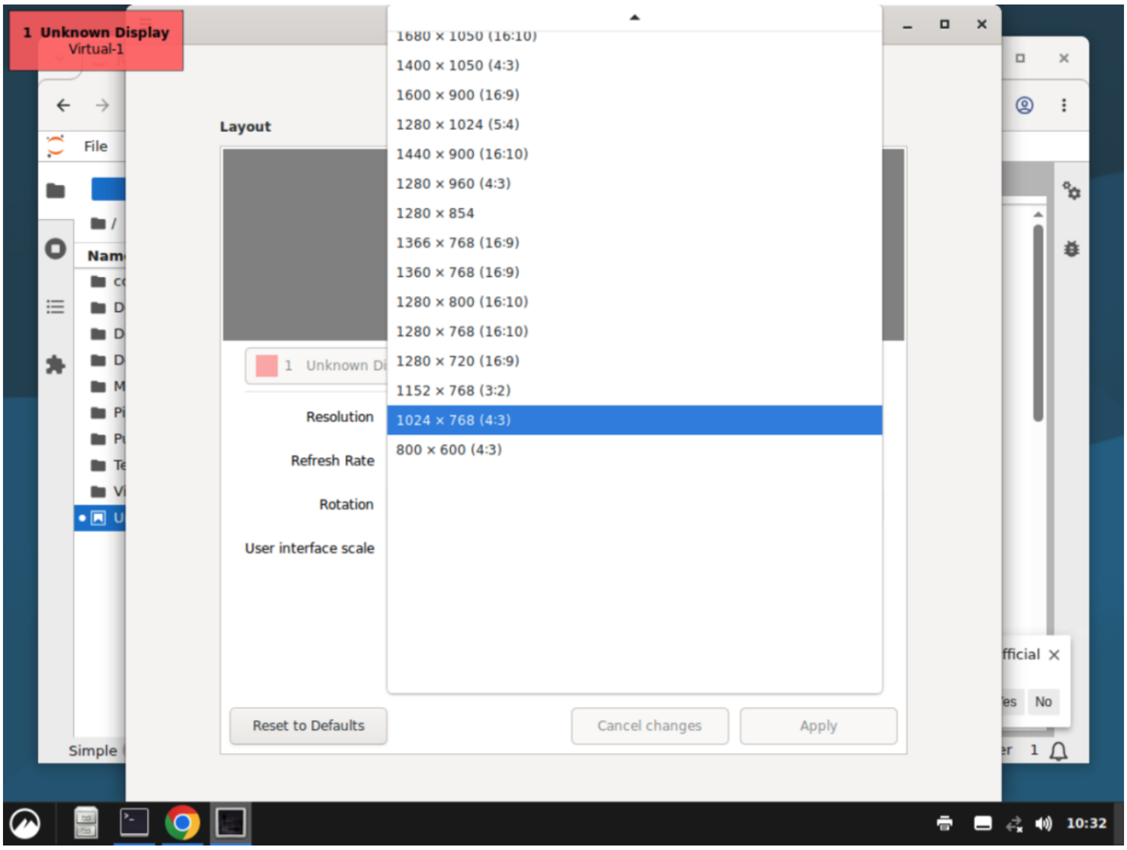
Website: www.parivartan.io | **Email:** connect@parivartan.io

Make the lab use your whole screen

Put your browser into **full-screen mode** — press **F11** on Windows or Linux, or **Ctrl + Cmd + F** on a Mac. The lab then fills your entire display. Press the same keys again to leave full-screen.

Change the screen resolution inside the lab

You can also set the lab desktop to a different resolution. Open the **applications menu** (bottom-left), type **Display**, and open **Display**. In the **Resolution** drop-down, pick a size that suits your screen — for example **1280 × 800** or **1366 × 768** — then click **Apply** and confirm **Keep changes**.



The Display settings — pick a size from the Resolution drop-down, then Apply.

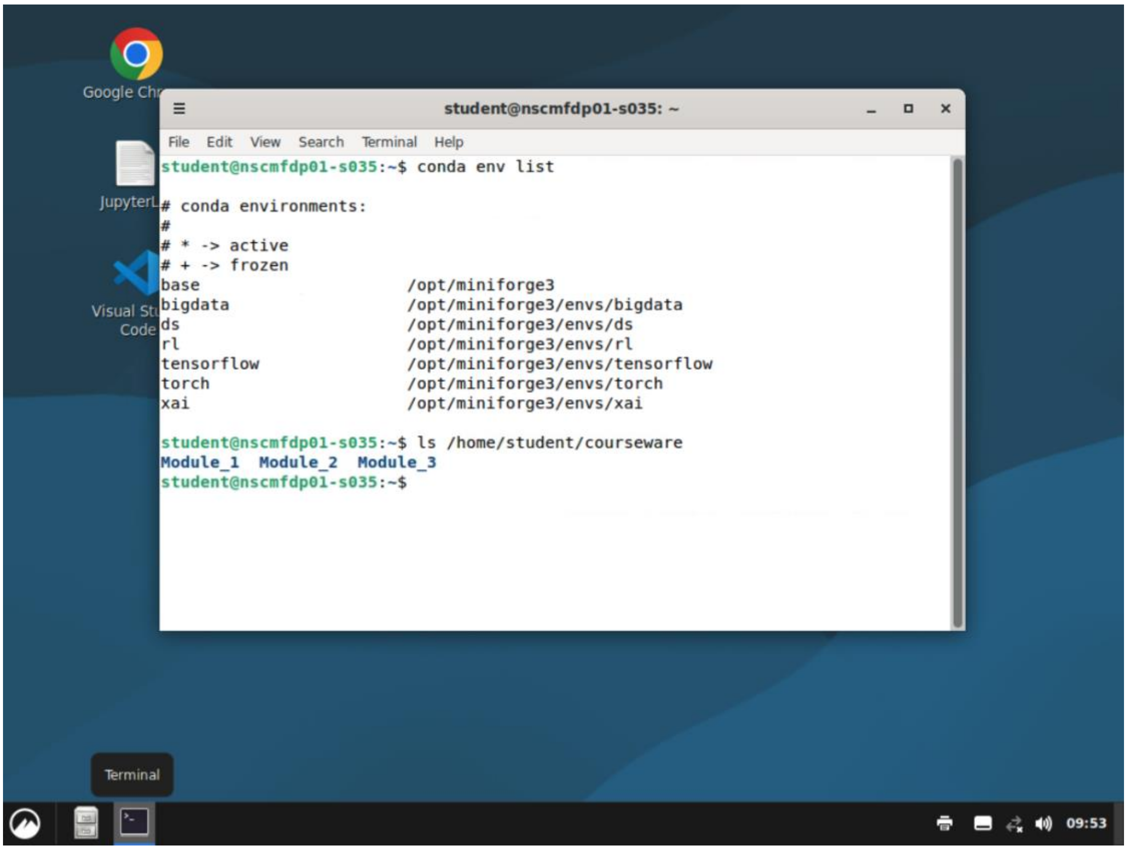
> **✓ Tip**
>
> A larger resolution gives you more room for windows but makes text smaller; a smaller resolution makes everything bigger. Choose whatever is most comfortable to read.

Step 7 — Python, conda environments and JupyterLab

Your lab uses **conda** (Miniforge, installed at /opt/miniforge3) to keep a separate, ready-made Python environment for each topic. Each environment has its own packages, and each is also registered as a **Jupyter kernel**, so you can open a notebook in exactly the environment you need.

To see them yourself, open the **Terminal** and run:

```
conda env list
```



conda env list and the courseware folder, run live inside the lab.

Your lab has these environments ready to use:

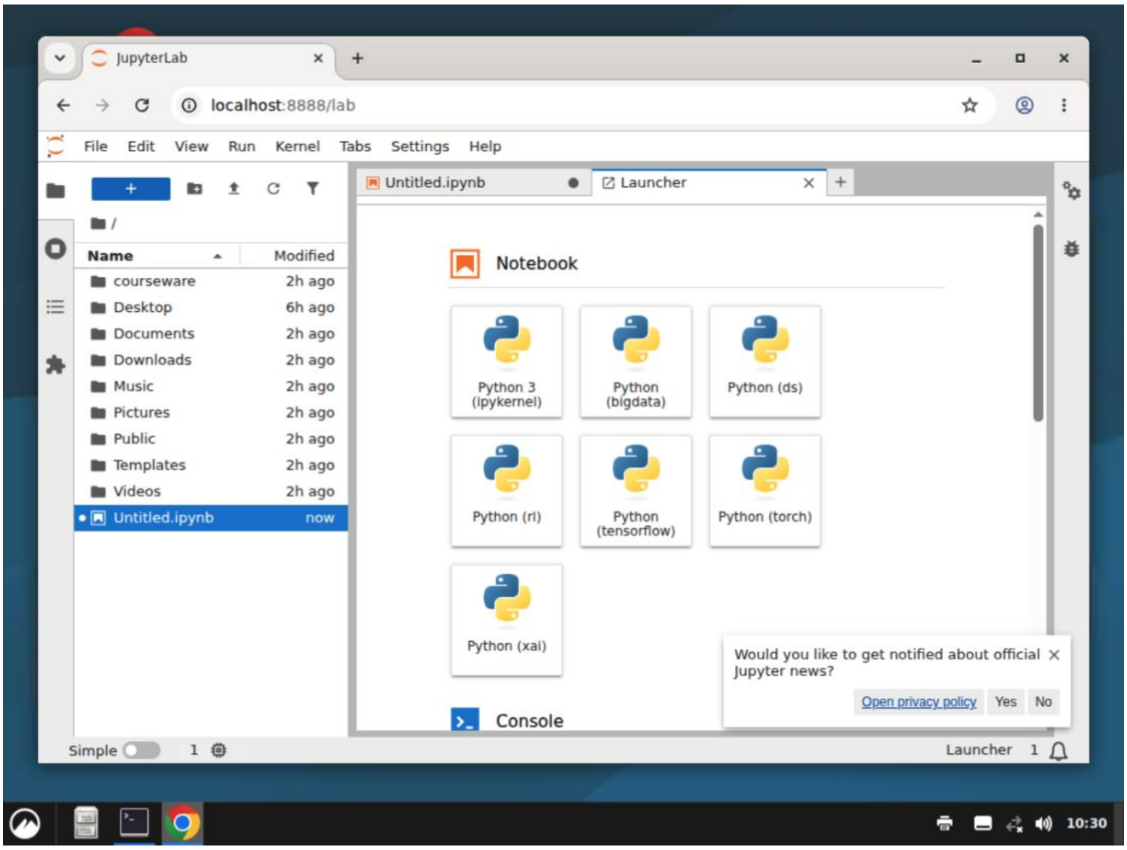
Environment	Use it for	Key packages included
ds	Data science & analytics (also hosts JupyterLab)	numpy, pandas, scipy, scikit-learn, matplotlib, seaborn, plotly, statsmodels, sympy, umap-learn, JupyterLab
torch	Deep learning with PyTorch	pytorch (CPU), torchvision, numpy, pandas, scikit-learn
tensorflow	Deep learning with TensorFlow	tensorflow-cpu, numpy, pandas, scikit-learn
bigdata	Big-data processing	pyspark + Java, pymongo, redis, kafka-python, dvc
xai	Explainable & fair AI	shap, lime, fairlearn, aif360, scikit-learn
rl	Reinforcement learning	gymnasium, stable-baselines3
base	The default conda environment	conda tooling (activate a topic environment above for your work)

> **A note on names**
> Deep-learning work is split into two environments — **torch** and **tensorflow** — so each stays clean and reliable. To switch environment in the Terminal, type, for example, `conda activate torch`.

Start JupyterLab

Double-click the **JupyterLab** icon on the desktop. It starts the JupyterLab server and opens it in Google Chrome inside the lab. (The first time, Chrome may ask a couple of welcome questions — choose **Stay signed out** / dismiss them.)

In the **Launcher** you will see one Notebook tile for **each** of your conda environments — **Python (ds)**, **Python (torch)**, **Python (tensorflow)**, **Python (bigdata)**, **Python (xai)** and **Python (rl)**. Click the one for the topic you are working on.



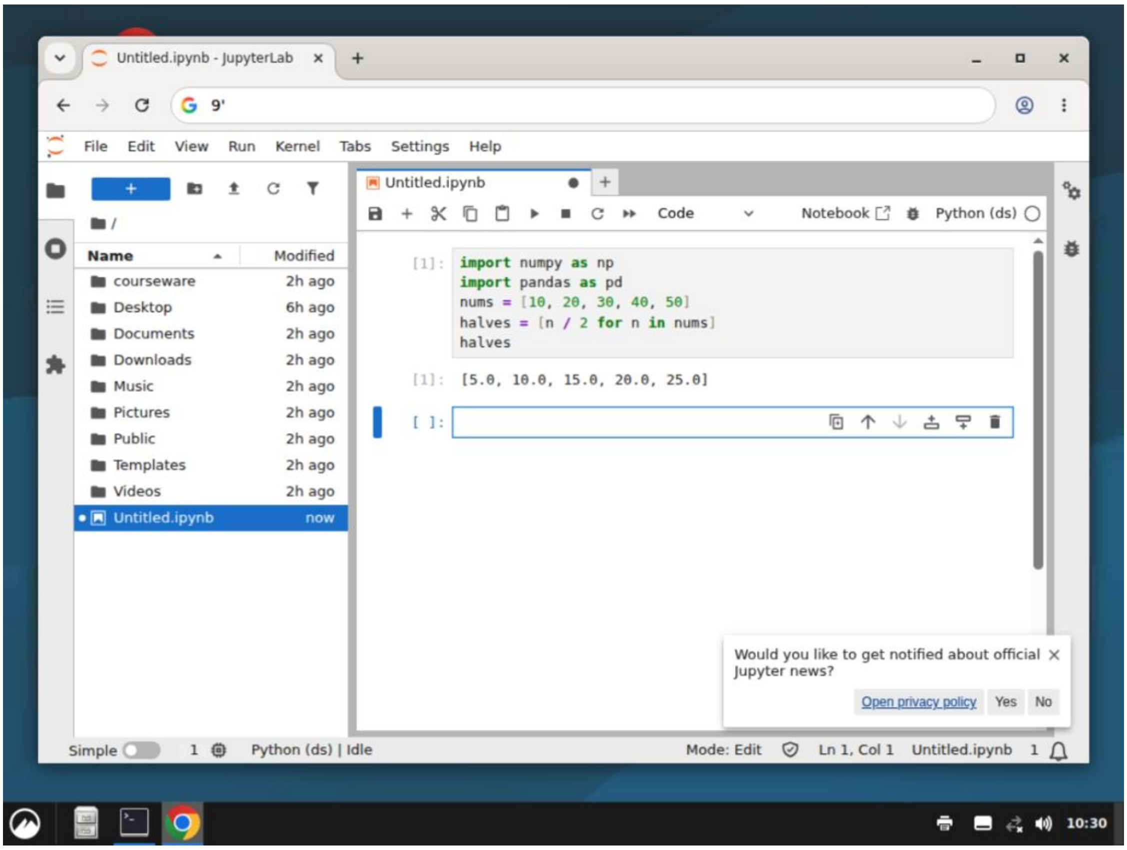
The JupyterLab Launcher — one notebook kernel for each conda environment.

Write and run your first code

A new, empty notebook opens. Click in the first cell, type some code, and press **Shift + Enter** to run it. The result appears just below the cell. For example:

```
import numpy as np
import pandas as pd
nums = [10, 20, 30, 40, 50]
halves = [n / 2 for n in nums]
halves
```

Running this imports the data-science libraries and prints the result `[5.0, 10.0, 15.0, 20.0, 25.0]` — confirming your environment works.

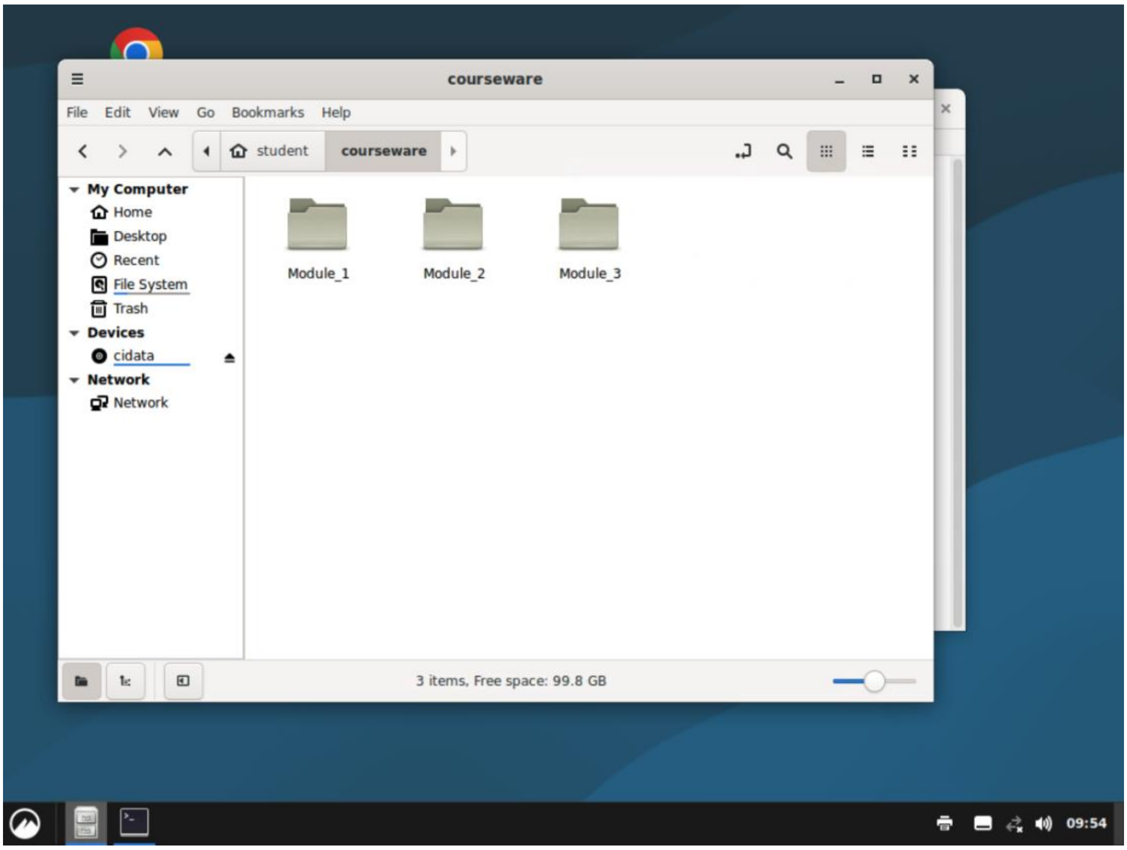


The notebook after running the cell — the output appears below the code.

Step 8 — Your courseware folder

Your course materials — slides, datasets, notebooks, exercises, and reference files — are kept in a folder called **courseware** inside your home directory.

To open it, use the **Files** application on the desktop and go to **Home → courseware**. The full path inside the lab is `/home/student/courseware`.



The courseware folder open in the Files app, organised by module.

The materials are organised by module — for example **Module_1**, **Module_2** and **Module_3** — and each module is further split into sub-topics (Module 1.1, 1.2, 1.3, and so on).

> **✓ Save your work inside the lab**
> Save the files you create inside your home folder (for example in `/home/student` or a sub-folder you make there). Work saved inside your environment stays available the next time you log in.

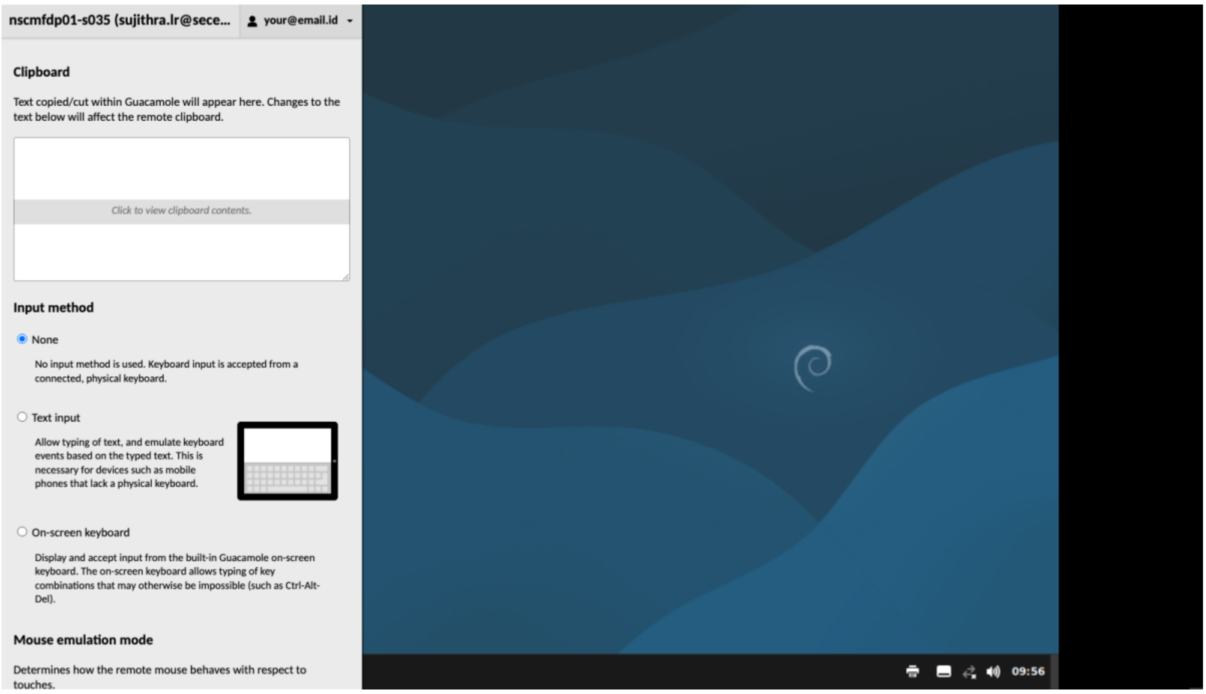
Step 9 — The in-lab menu (Ctrl + Alt + Shift)

While you are inside a lab, there is a hidden control menu you can open at any time. Press these three keys together:

Ctrl + **Alt** + **Shift**

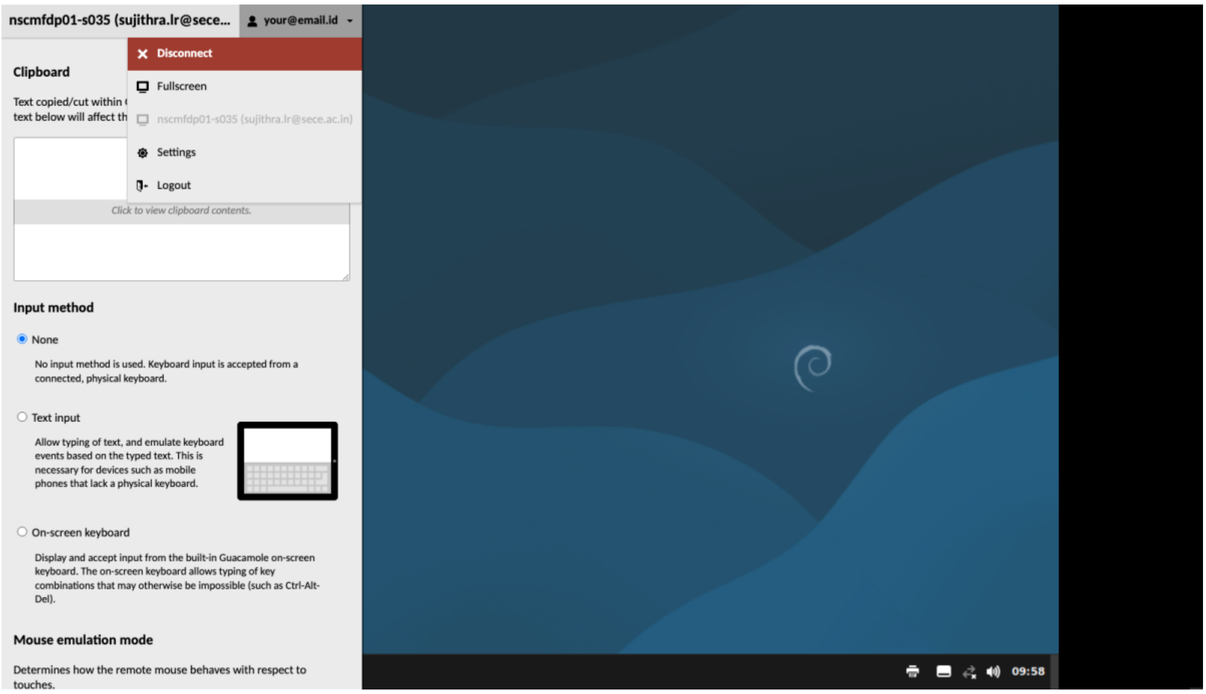
A panel slides in from the side of the screen. From here you can:

- **Copy and paste between your own computer and the lab** — use the shared **Clipboard** box. Paste text into it to make it available inside the lab, or copy text out of the lab to your own machine.
- **Change input and mouse settings** — handy for touch devices or an on-screen keyboard.
- **Disconnect** from the lab when you are finished (see Step 10).



The Ctrl + Alt + Shift menu, with the shared Clipboard box at the top.

The **Disconnect** option is in the menu at the top — click your email address to open it, as shown below.



The user menu, showing Disconnect, Fullscreen, Settings and Logout.

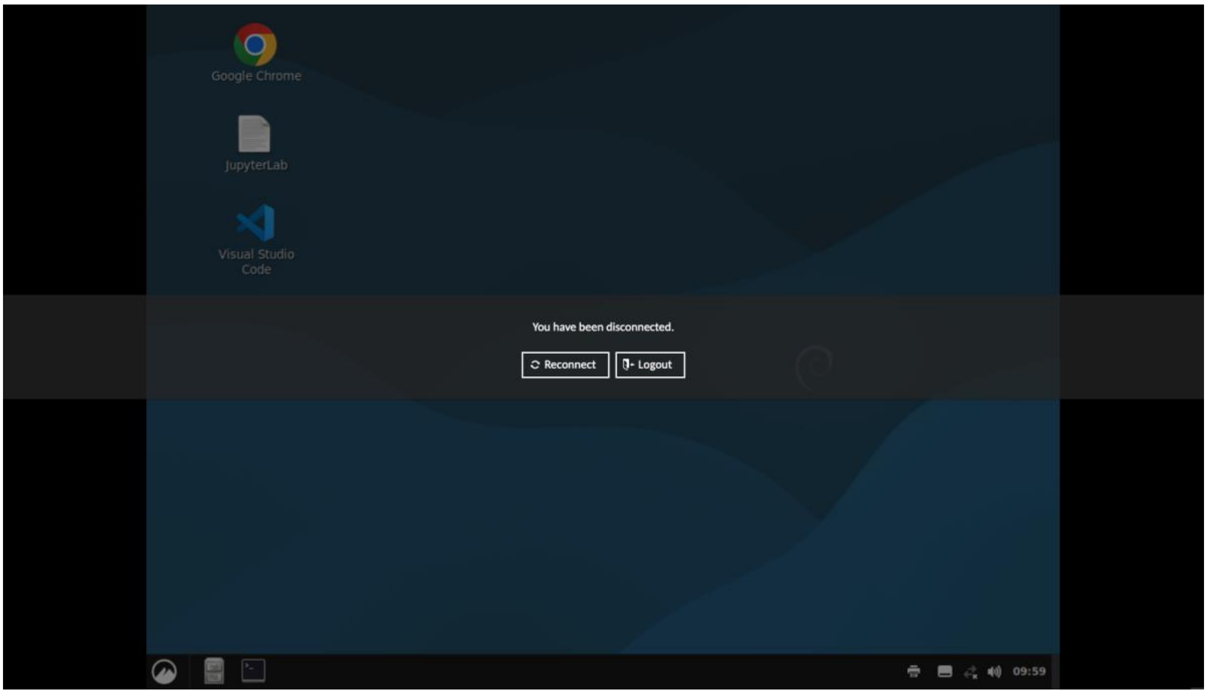
To **close the menu and return to your desktop**, simply press **Ctrl + Alt + Shift** again.

Why a special shortcut?
Because the lab uses your whole screen, copy-paste between your computer and the lab goes through this Clipboard panel. **Ctrl + Alt + Shift** is the one shortcut to remember — it opens and closes this menu.

Step 10 — Disconnecting safely

When you have finished a session:

1. **Save your work** inside the lab first (in your home folder).
2. Open the menu with **Ctrl + Alt + Shift**.
3. Click your email address, then click **Disconnect**.
4. You will see a "You have been disconnected" screen. From here you can **Reconnect** to return to your lab, or click **Logout** to sign out fully.



The disconnected screen — Reconnect to continue later, or Logout to sign out.

✓ Your environment keeps running
Disconnecting only closes *your view* of the lab — your saved files and setup remain intact. When you log back in, you will pick up where you left off.

⚠ On shared or public computers
Always click **Logout** (not just close the browser tab) when you are using a computer that is not your own, so that no one else can open your lab.

Tips and troubleshooting

If this happens…	Try this
The sign-in page will not load	Check your internet connection and that the address is exactly https://labs.learningfootprints.com. Refresh the page.
"Invalid Login" message	Re-type your email and password carefully (passwords are case-sensitive). If it still fails after a few tries, contact your coordinator — repeated wrong attempts can temporarily block sign-in for security.
The lab screen is frozen or grey	Open **Ctrl + Alt + Shift**, click Disconnect, then Reconnect (or open the environment again).
Copy-paste is not working	Use the **Clipboard** box in the **Ctrl + Alt + Shift** menu to move text in and out of the lab.
The desktop is too small or too large	Put your browser in full-screen mode (F11), then reconnect so the lab matches your screen size.
The mouse pointer feels offset or laggy	This usually settles after you reconnect. A stable, wired or strong Wi-Fi connection gives the smoothest experience.
I cannot find my course materials	Open the **Files** app and go to **Home → courseware** (/home/student/courseware).
"Command jupyter not found" or wrong packages	Activate the right environment first, e.g. `conda activate ds`, or launch JupyterLab from its desktop icon.

Getting help

If you hit a problem this guide does not cover, please reach out to your **program coordinator or trainer**. It helps if you mention:

- The email address you log in with.
- The name of the environment you were trying to use.
- What you were doing when the problem happened, and any message you saw.

> **Quick reference**
> **Website:** https://labs.learningfootprints.com
> **Open / close the in-lab menu:** **Ctrl + Alt + Shift**
> **Course materials:** Files → Home → courseware
> **Switch Python environment:** `conda activate <name>`
> **Finish a session:** Save → Ctrl + Alt + Shift → Disconnect → Logout